

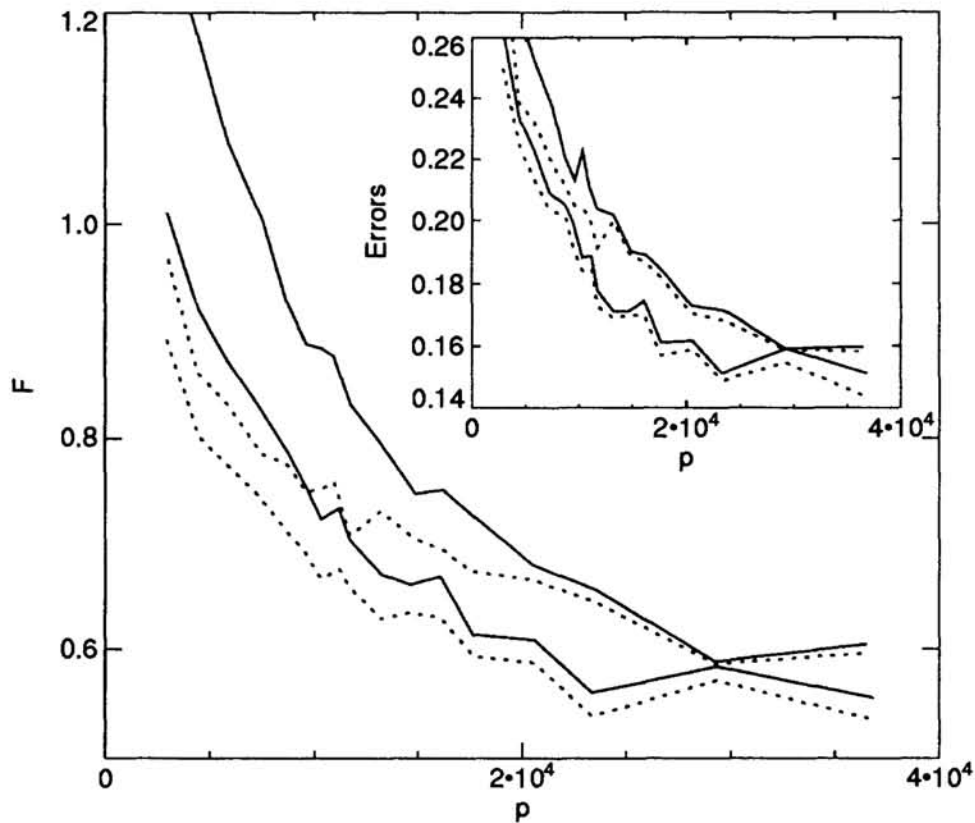


Figure 2: The top full line corresponds to the generalization error after 300 epochs (300 cycles through the training set) without a weight decay. The lower full line is with a weight decay. The top dotted line is the lowest error seen during learning without a weight decay, and the lower dotted with a weight decay. The size of the weight decay was $\lambda = 0.00008$.

Insert: Same figure except that the error rate is shown instead of the squared error. The error rate is the fraction of wrong phonemes when the phoneme vector with the smallest angle to the actual output is chosen, see [11].

Results are shown in fig. 2. There is a clear improvement in generalization error when weight decay is used. There is also an improvement in error rate (insert of fig. 2), but it is less pronounced in terms of relative improvement. Results shown here are for a weight decay of $\lambda = 0.00008$. The values 0.00005 and 0.0001 was also tried and gave basically the same curves.

7 CONCLUSION

It was shown how a weight decay can improve generalization in two ways: 1) It suppresses any irrelevant components of the weight vector by choosing the smallest vector that solves the learning problem. 2) If the size is chosen right, a weight decay can suppress some of the effect of static noise on the targets. Static noise on the targets can be viewed as a model of learning an unrealizable function. The analysis assumed that the network could be expanded around an optimal weight vector, and

therefore it is strictly only valid in a little neighborhood around that vector.

The improvement from a weight decay was also tested by simulations. For the NetTalk data it was shown that a weight decay can decrease the generalization error (squared error) and also, although less significantly, the actual mistake rate of the network when the phoneme closest to the output is chosen.

Acknowledgements

AK acknowledges support from the Danish Natural Science Council and the Danish Technical Research Council through the Computational Neural Network Center (CONNECT).

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